

Self-Supervised Methods for Radio Propagation Signals

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Physical Twin description

Any device connected by wireless to a radio access network (5G/WiFi...) or to a set of receivers (bluetooth, ADS-B...)

Digital Twin usages

- Real-time pseudo-location tracking
- Radio resource management
- Predictive networking resource allocation
- Authentication

Digital Twin Description

The digital twin is the mobile wireless device. We consider in particular its position as the main property, together with other features (device features, typical mobility patterns, etc.)

Engineering Digital Twin challenges

- Multi-actor fusion and alignment: multiple channel charts produced by different devices in a network are not necessarily spatially consistent; the corresponding digital twin representations require specific alignment methods with limited information exchange.
- Anomaly detection:
- Achieving temporal consistency in a non-stationary world: in a self-supervised learning setting, it is difficult to reliably discriminate the non-stationary effects (due to the physical world being in constant evolution) from the stationary effects which can be statistically inferred. Inclusion of expert knowledge is required to resolve this stability-plasticity dilemma.

